Definition of a Correction Ratio approach for Fission Matrix Temperature Interpolation and Monte Carlo Uncertainty Propagation Methods for Results Assessment

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Abstract

The fission matrix (FM) method enables whole-core reactor transport calculations within seconds by interpolating a precalculated database of Monte Carlo (MC)–tallied coefficients across different operational parameters instead of running a direct MC calculation for every configuration of interest. This work focuses on the parametrization of the FM with respect to the temperature distribution within a micro core. Interpolation implicitly assumes that each coefficient depends only on the local temperature of cells where fission occurs. This assumption breaks down under large, spatially sharp temperature gradients. In this work, a Core Temperature Correction Ratio (CTCR) is introduced to compensate this effect.

The uncertainty quantification of the results is necessary. The uncertainty budget is made of: the reference case's MC statistics, and the uncertainty of the eigenvalue (k_{eff}) and eigenvector ($\nu\Sigma_f\phi$) results of the *interpolated* matrix itself: a new object, assembled from the database's own finite-statistics matrices, which carries no MC replicas of its own. We construct the latter by resampling the database and recomputing the interpolated (and corrected) result many times, comparing an uncorrelated baseline, an innovative derivation of a analytical model, a measured-correlation model, and a genuine brute-force reconstruction from 150 independent replicas. Validated against brute force across 12 test cases, the analytical model reproduces the reference k_{eff} uncertainty to within a mean absolute deviation of 4.6%, matching or exceeding the two more expensive alternatives, while requiring no MC replicas beyond the nominal database already needed for interpolation. Combining both uncertainty sources, the benefits of the CTCR are evident in almost all cases. In general, it removes most of the point-estimate k_{eff} discrepancy for sharply varying, highly localized temperature fields, whereas the improvement becomes less pronounced (and less necessary) for smoother temperature distributions. The local fission-source error follows the same pattern, improving significantly wherever a strong, spatially localized temperature gradient is present; where the correction succeeds, the residual discrepancy is generally consistent with the combined statistical noise of the reference case and the interpolated database.

Keywords — Fission matrix, correction ratio, uncertainty quantification, covariance, HPMR, temperature

I. INTRODUCTION

Monte Carlo (MC) methods provide the high-fidelity treatment of the neutron transport equation for neutrons in reactor physics applications, at the cost of a statistical uncertainty that decreases only as the inverse square root of the simulated particle histories. Hybrid neutron transport methods based on the fission matrix (FM) formulation [1] address this cost by precalculating a database of FM coefficients under a set of reference system configurations and then interpolating and combining these coefficients to represent an arbitrary configuration within the database's validity range.

There are two different strategies for obtaining the FM of a single core configuration. The first one assembles the whole-core FM by running one fixed-source calculations for each "fundamental cell" of the system. As well described in [2], the fundamental units are identified by exploiting the geometrical and material similarities between repeated units to limit the required database calculations. Reflectors and structural materials are deliberately excluded from the FM tally to preserve this similarity, and their influence is recovered afterward through dedicated boundary and discontinuity corrections. This method is the approach adopted in RAPID [2], which is a mature implementation of this approach, yielding whole-core transport solutions within a few minutes at accuracy comparable to the underlying MC calculations, aided by dedicated techniques for FM homogenization [3], temperature interpolation [4], and correction of material and control-rod discontinuity effects [5, 6, 7]. The present work adopts the second strategy to obtain the whole-core FM for a single configuration, which consists in tallying the FM elements directly from a single whole-core criticality calculation in Serpent [8], so that reflectors, structural materials, and inter-assembly heterogeneity effects are already embedded in each coefficient, and a complete fission matrix is obtained from one calculation rather than assembled from fundamental-unit responses. The selection of the most suitable approach is case-specific: it depends on the reactor design and of purpose of the analysis keeping in mind the features of the two methods. The method used in RAPID, in fact, requires, where applicable, the recognition of the fundamental cells of the system, and the definition of multiple fixed-source MC calculations to build up one single whole-core FM. Additionally, to guarantee the right convergence of the eigen-problem's results of the resulting FM, it is important that the mesh of the single fixed-source cases is fine enough to capture all the local effects, at the same time a too small grid would be a burden memory-wise. This means that the

dimension of the FM is not completely arbitrary but it must be a trade-off between the result-accuracy and the hard-disk storing capacity. Generally speaking the fixed-source calculations need less computational time and the uncertainty of the tallied FM elements is small if enough neutron histories are set. The whole-core FM deriving from a criticality calculation, instead, uses a already-converged neutron source, which means that there is not direct dependence of the fundamental eigenvalue and eigenfunction accuracy. By accumulating the fission matrix coefficients on the fly, the within-cell spatial distributions of the fission source are much more accurate than the fixed-source technique, which allows for much larger spatial cells to be used. On the other hand, to obtain data with a low-enough uncertainty, especially for pin-wise cells a high number of MC histories are necessary.

The selection of the second method is justified form the fact that the goal is to define an easy and straightforward workflow to obtain reliable results based on the FM method for an online core-monitoring tool for the Westinghouse eVinci reactor, without the need the have a pin-wise accuracy.

The system studied here is a modified Heat-Pipe Micro-Reactor (HPMR) concept, and the operational variable of interest is the mesh cell temperature: the FM database is tallied at a set of uniform reference temperatures, and an arbitrary in-core temperature field is represented by interpolating each coefficient across that database. As a general approach widely adopted to deal with interpolations performed on temperature distributions, it is assumed that a coefficient a_{ij} depends only on the *local* temperature of the destination cell, i.e. the i-th cell. However, under a large, spatially sharp temperature gradient that assumption breaks down and the systematic error affecting the interpolated fission matrix can increase. To compensate it, we introduce a Core Temperature Correction Ratio (CTCR), calibrated once from two additional FMs, and show that it reduces the point-estimate discrepancy between interpolated and directly-simulated reference results across a range of test cases.

To better understand the systematic error and the improvement provided by the CTCR, a point-estimate comparison of the interpolated FMs and the reference one directly calculated with Serpent could not be enough to distinguish a real, uncorrected modeling bias, since both the database and the reference case are generated by Monte Carlo simulations, which always exhibit intrinsic statistical noise. Making that distinction requires quantifying the statical uncertainty

on both quantities being compared. This estimation is straightforward for the reference case: Serpent [8] naturally reports its own statistical uncertainty for each tallied quantity, including both k_{eff} and the fission source, F. On the contrary, the results of the interpolated matrix are provided with any uncertainty: the eigenvalue and eigenvector of the matrix produced by interpolation (and CTCR correction) are new objects, and unlike a direct MC run it has no replicas of their own from which an uncertainty could be read off. The second aim of this work is centered on this practical gap, i.e. constructing a computationally smart uncertainty estimate for the interpolated FM. This will allow us to better quantify any systematic error and the improvement provided by the correction ratio. In the following sections a detailed description of the database construction and the interpolation pipeline are described with the CTCR definition; the combined uncertainty source (reference and interpolated cases) problem is described; the four approaches considered for estimating the database-side contribution, and the error metrics adopted throughout. Finally, the reference test cases are described and the corresponding error metrics are reported.

II. SYSTEM GEOMETRY DESCRIPTION

The system considered in this work is a modified version of the Heat-Pipe Micro-Reactor (HPMR) concept developed at Argonne National Laboratory under the NEAMS Micro-Reactor Application Drivers program [9]. It consists of 127 hexagonal, TRISO-fueled assemblies embedded in a graphite monolith and surrounded by a beryllium radial reflector. Control rods are removed from the configuration, and the core is treated as axially infinite, so that the analysis is carried out on a two-dimensional radial model throughout. The system is shown in Figure 1.

III. METHODOLOGY

III.A. The Fission Matrix Method

The fission matrix (FM) method [1] recasts the k-eigenvalue transport problem into a compact linear-algebraic form by partitioning the fissile domain into N spatial regions and describing the transport process among them over a averaged single neutron generation. Applying the integral form of the k-eigenvalue transport equation over the volumes of a source region j and a destination

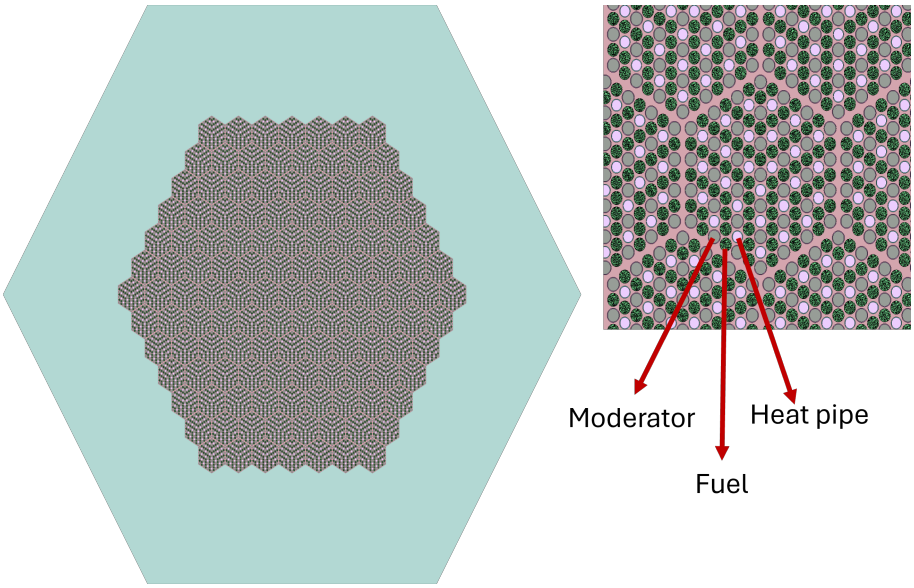

Fig. 1. System geometry.

region i yields an algebraic eigenvalue problem,

$$F_i = \frac{1}{k} \sum_{j=1}^{N} a_{ij} F_j, \qquad \Longleftrightarrow \qquad \mathbf{A}\,\mathbf{F} = k\,\mathbf{F}, \tag{1}$$

where F_j is the fission-neutron source integrated over region j, and the coefficient a_{ij} encodes the transport process yielding the expected number of fission neutrons born in region i per fission neutron emitted in region j. The $N \times N$ array of these coefficients is the fission matrix $\mathbf{A}$: its dominant eigenvalue is k_{eff} and the associated eigenvector is the region-wise fission-source distribution [1].

Since each coefficient is defined per emitted source neutron, the fission multiplicity is already embedded in it: a source neutron that induces a fission contributes $\bar{\nu}$ neutrons to the destination region. A key practical feature of the method is that the coefficients can be accumulated at negligible additional cost during an ordinary Monte Carlo criticality calculation, by tallying, over the active cycles, the region of birth of each fission neutron against the region of birth of its parent. In this work the full fission matrix (one region per fuel assembly, yielding a 127×127 operator) is obtained in this way from a single whole-core Serpent calculation. Hence, every coefficient reflects the complete core environment.

III.B. Database, Interpolation, and Correction Ratio

Rather than tallying a fission matrix directly for every temperature field of interest, the method precalculates a database of fission matrices at a set of uniform reference temperatures and interpolates within it. To keep the methodology practical and computationally attractive, it is important to maintain good accuracy while relying on a compact and simply defined database. For this reason, the database considered here contains the coefficients a_{ij} at only eight uniform temperatures spanning 300–1000 K in 100 K increments. For an arbitrary temperature field, each coefficient is first obtained by piecewise-linear interpolation between the two bracketing database temperatures:

$$a_{ij}(T_i) = a_{ij}^{-} + \frac{T_i - T_i^{-}}{T_i^{+} - T_i^{-}} \left(a_{ij}^{+} - a_{ij}^{-}\right), \tag{2}$$

where a_{ij}^{-} and a_{ij}^{+} are the values of a_{ij} at the database temperatures immediately below and above the arbitrary destination-cell temperature T_i.

Eq. 2 treats each coefficient as depending only on the local temperatures i, and therefore misses the influence of the surrounding cells. Under a large, spatially sharp gradient this is precisely where the interpolated matrix acquires a systematic error. The Core Temperature Correction Ratio (CTCR) defined in this work tries to compensate this by informing the numerical value of a generic matrix element a_{ij}, based on the couple of temperatures (T_i, T_j) through a correction ratio $\hat{r}_{ij}$. This correction is calibrated from two additional, directly simulated FMs, defined with a "Dirac temperature distributions" shown in Figure 2.

Because the correction must handle both a locally *hot* cell embedded in a colder surrounding and the reverse case, both Dirac FMs of Figure 2 are used: one with the central assembly C driven to the hottest database temperature (1000 K) while the rest of the core sits at the coldest temperature (300 K), and one with the assignment reversed. Comparing each Dirac FM against its own locally-interpolated counterpart gives two ratios, each a function of the geometric distance d between C and a generic cell j, including $j = C$ itself ($d = 0$) (exploiting the hexagonal core's sextant symmetry, so that only d, not direction, matters),

$$r^{+}(d) = \frac{a_{Cj}^{\mathrm{dirac},+}}{a_{Cj}^{\mathrm{interp}}}, \qquad r^{-}(d) = \frac{a_{Cj}^{\mathrm{dirac},-}}{a_{Cj}^{\mathrm{interp}}}, \tag{3}$$

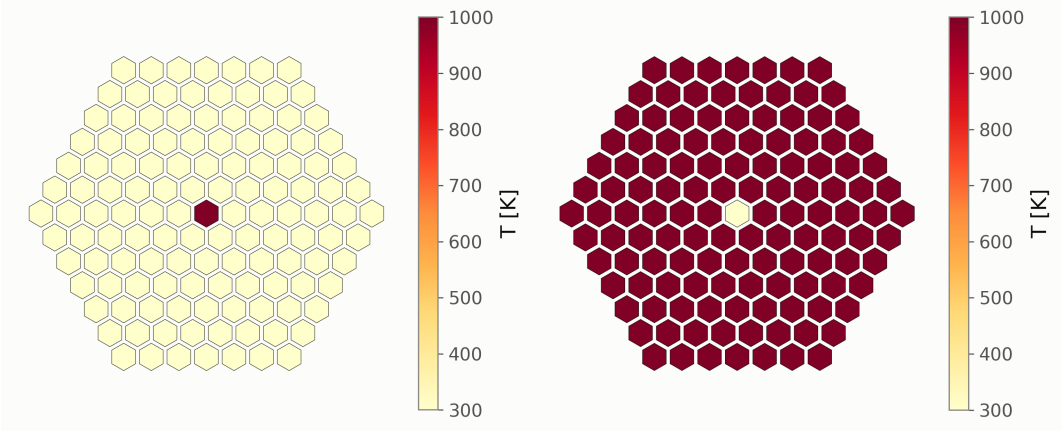

Fig. 2. The two additional cases whit "Dirac temperature distributions", used to calibrate the CTCR.

where the superscript $+$ $(-)$ denotes the hot-center (cold-center) Dirac case, and a_{Cj}^{interp} is the coefficient the ordinary interpolation of Eq. (2) would give for that same, uniform-elsewhere temperature field. Each ratio in isolation quantifies how much the local-temperature interpolation assumption misses, in the most extreme, the most spatially sharp case the database can represent; combined, the two make the correction able to handle a temperature difference of either sign.

For an arbitrary temperature field, the correction applied to a generic interpolated coefficient a_{kl}^{interp} is

$$\hat{r}_{kl}(\Delta T_{kl}, d) = \frac{\Delta T_{kl}}{\Delta T_{\text{dirac}}} \left(r^{\pm}(d) - 1\right) + 1, \qquad \Delta T_{kl} = \begin{cases} T_k - T_l, & k \neq l, \\ T_k - \dfrac{1}{|\mathcal{N}(k)|} \displaystyle\sum_{n \in \mathcal{N}(k)} T_n, & k = l, \end{cases} \tag{4}$$

where $\Delta T_{\text{dirac}} = 700$ K is the calibration temperature span common to both Dirac cases, and $r^{+}(d)$ is used when $\Delta T_{kl} > 0$ (cell k hotter than l, or than its own first-ring neighborhood average $\mathcal{N}(k)$ when $k = l$) while $r^{-}(d)$ is used when $\Delta T_{kl} < 0$ — i.e., the calibration case matching the *sign* of the local temperature difference, evaluated at its own geometric distance d. For a diagonal coefficient ($k = l$, $d = 0$) this reuses $r^{\pm}(0)$ from the same Dirac calibration of Eq. (3), so no additional simulation or calibration data is needed to cover this case. In other words, the logic leading Eq. (4), is to transpose and scale the the coefficients of $r^{+}(d)$ (and $r^{-}(d)$) of Eq. (3), for any couple of cells of the core with any temperature distribution. If $k \neq l$ the scaling (or weighting)

coefficient is proportional to the actual temperature difference between cell l and cell k, divided by the temperature difference of the central cell and the peripheral ones of the calibration cases. When $k = l$, instead, source and destination cells correspond, thus the correction coefficients of Eq. (3), carries a different information: how much the raw interpolation mistakes the prediction of the FM element a_{kk}, when it is surrounded by $\mathcal{N}(k)$ neighbors with a total average temperature of $\frac{1}{\mathcal{N}(k)} \sum_{n \in \mathcal{N}(k)} T_n$. Thus the weighting coefficient in Eq. (4) is defined consequently.

In this work the two variants compared are referred to as *raw* (interpolation only) and *corrected* (interpolation plus CTCR).

III.C. Two Sources of Uncertainty for k_{eff} and F

Comparing an interpolated (and CTCR-corrected) results (here generally indicated with X) to a direct reference calculation involves two statistically independent Monte Carlo noise sources, which combine in quadrature:

$$\sigma^2_{X,\text{tot}} = \sigma^2_{X,\text{ref}} + \sigma^2_{X,\text{interp}}. \tag{5}$$

(a) *Reference-case noise*, $\sigma_{X,\text{ref}}$: the statistical uncertainty Serpent reports directly for the single reference simulation being compared against.

(b) *Interpolated noise*, $\sigma_{X,\text{interp}}$: the noise that the result X inherited by the interpolated (and corrected) matrix from the 8 database calculations it is built from. This uncertainty source is not available like for (a): it has to be estimated.

III.D. Quantification of $\sigma_{X,\text{interp}}$

In this work four different approaches are introduced to propagate the database uncertainty to the interpolated FM results. Three of these methods share the same approach: the nominal eight-temperature database is perturbed with random noise, the interpolator and correction of Eq. (4) are reapplied, and the procedure is repeated many times. The spread of the resulting $k_{\text{eff}}^{\text{interp}}$ and fission source F^{interp} over these repetitions is the database-side uncertainty.

The four approaches differ in how the random perturbation is enforced. Indicating the perturbation of the retained coefficients as a vector $\boldsymbol{\delta}$ drawn from a zero-mean multivariate normal

distribution, $\boldsymbol{\delta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, the covariance matrix $\boldsymbol{\Sigma}$ factors as

$$\boldsymbol{\Sigma} = \mathbf{D}\,\boldsymbol{\rho}\,\mathbf{D}, \tag{6}$$

where $\mathbf{D}$ is the diagonal matrix of the per-coefficient standard deviations reported by Serpent, and $\boldsymbol{\rho}$ is the correlation matrix. The amplitude $\mathbf{D}$ is common to all approaches — it is read directly from the nominal database — so the four differ only in what they assume for $\boldsymbol{\rho}$: from neglecting correlation entirely, to predicting it analytically, to measuring it, to bypassing the construction altogether.

Operationally, the perturbation is drawn column by column. Because the covariance structure derived in Sec. III.E couples only coefficients sharing a source column, $\boldsymbol{\Sigma}$ is block diagonal, with one $N \times N$ block per source region j. Factorizing the dimensionless correlation block as $\boldsymbol{\rho}_j = \mathbf{C}_j \mathbf{C}_j^{\top}$ by Cholesky decomposition, the m-th realization of the perturbation of column j at database temperature T is

$$\boldsymbol{\delta}_{:,j}^{(m)}(T) = \mathbf{D}_j(T)\,\mathbf{C}_j(T)\,\mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N), \tag{7}$$

so that $\mathrm{Cov}(\boldsymbol{\delta}_{:,j}) = \mathbf{D}_j \boldsymbol{\rho}_j \mathbf{D}_j$ by construction. A single vector $\mathbf{z}$ is shared by all coefficients of the same column, which is what imposes the correlation; the marginal standard deviation of each individual coefficient remains σ_{ij} regardless of $\boldsymbol{\rho}$. The factorizations are performed once and reused across all realizations. Since the eight database temperatures are independent Monte Carlo calculations, their perturbations are drawn independently, and no inter-temperature covariance is modeled.

Each perturbed database is then passed through the full pipeline: interpolation of Eq. (2), correction of Eq. (4), and solution of the eigenproblem of Eq. (1), yielding one realization of $k_{\mathrm{eff}}^{\mathrm{interp}}$ and of the normalized fission source. The procedure is repeated $M = 500$ times, and σ_{interp} is taken as the standard deviation of the resulting ensemble. Propagating the database covariance through a linearized sensitivity expression would be cheaper, but the CTCR of Eq. (4) is a nonlinear function of the database coefficients, which appear in both the numerator and the denominator of the calibration ratios of Eq. (3); resampling the full pipeline avoids any linearization of this dependence.

(i) **Uncorrelated resampling ($\boldsymbol{\rho} = \mathbf{I}$).** Each retained coefficient is perturbed independently using only its own MC-reported variance, $a_{ij} \to a_{ij} + \delta_{ij}$ with $\delta_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$. This is equivalent to setting $\boldsymbol{\rho}$ to the identity matrix and is the simplified assumption adopted in prior FM uncertainty-quantification work [10]. It serves as the natural baseline against which to evaluate the impact of modeling correlation.

(ii) **Analytical model ($\boldsymbol{\rho} = \boldsymbol{\rho}_{\mathbf{pred}}$).** The correlation matrix is predicted in closed form, $\boldsymbol{\rho}_{\mathrm{pred}}$, derived directly from the branching structure of the fission process. To the best of our knowledge, this closed-form prediction of the fission-matrix correlation structure from the branching process, and its use to build the interpolated matrix uncertainty, has not been reported before: prior FM uncertainty quantification works either neglect inter-coefficient correlation or estimate it empirically. The approach requires no Monte Carlo replicas beyond the single nominal database already needed for interpolation.

(iii) **Correlated resampling ($\boldsymbol{\rho} = \hat{\boldsymbol{\rho}}$).** The correlation matrix is instead obtained from the cycle-by-cycle tallies of a single MC run, treating each active cycle as a pseudo-replica of the same underlying random vector. Since Serpent does not provide the fission matrix on a per-cycle basis, this required a modification of the source code to tally and print each active cycle's fission matrix separately.

This $\hat{\boldsymbol{\rho}}$ is measured once, at 300 K, and reused unchanged at every other database temperature. An analysis of the reasonableness of this approach is also performed in the following.

(iv) **Brute force.** The full eight-temperature database is rebuilt from scratch $N = 150$ times, with one genuinely independent MC replica per temperature and per repetition, reapplying the interpolator and correction each time. This bypasses the $\boldsymbol{\Sigma} = \mathbf{D}\boldsymbol{\rho}\mathbf{D}$ construction entirely: all correlations are present in the replicas by construction, with no modeling assumption. It is the most direct and most expensive estimate, and serves as the reference against which approaches (i)–(iii) are verified.

III.E. Deriving the Analytical Correlation Model

Coefficients tallied within a single Monte Carlo criticality calculation are not, in general, statistically independent, because the number of neutron histories simulated per active cycle is

fixed, a fact responsible for well-documented biases and correlations in cycle-based Monte Carlo estimators [11]. For two FM coefficients sharing the same source column j (i.e., a_{ij} and a_{lj}, $i \neq l$), this mechanism can be made explicit and turned into a closed-form prediction.

At the start of a cycle, the number of source neutrons born in column j, N_j, is fixed and inherited from the fission bank of the previous cycle. Each of these N_j histories picks, independently of the others, exactly one out of two mutually exclusive outcomes: it either causes fission in some destination cell i with probability π_i, or it causes fission in another cell, with the complementary probability $1 - \pi_i$. Hence, the process follows a Bernoulli distribution.

If it does cause fission, all $\bar{\nu}$ secondaries are born together at that one destination. Writing the contribution of history s to destination i as $X_s^{(i)} = \bar{\nu}\, I_s(i)$, where $I_s(i)$ is 1 if history s fissions in i and 0 otherwise, the retained coefficient is the per-source-neutron average:

$$a_{ij} = \frac{1}{N_j} \sum_{s=1}^{N_j} X_s^{(i)}. \tag{8}$$

The tracking used here is analog: no variance reduction is applied, and a history is terminated when it causes fission. A single history therefore contributes to one destination only, so $I_s(i) \cdot I_s(l) = 0$ for $i \neq l$, and different histories are independent of one another (linear transport: simulated particles do not interact). Under implicit capture, where a history survives absorption and may fission more than once, or under particle splitting, where the descendants of one source neutron are not independent, these two properties would have to be revisited. Hence, for a single history,

$$\mathrm{Var}\left(X_s^{(i)}\right) = \bar{\nu}^2\, \pi_i(1-\pi_i), \qquad \mathrm{Cov}\left(X_s^{(i)}, X_s^{(l)}\right) = \bar{\nu}^2 \left[\underbrace{E\big(I_s(i) I_s(l)\big)}_{=0} - \pi_i \pi_l\right] = -\bar{\nu}^2\, \pi_i \pi_l, \tag{9}$$

and summing N_j independent, identically distributed histories (thus summing over s) and dividing by N_j as in Eq. (8),

$$\mathrm{Var}(a_{ij}) = \frac{\bar{\nu}^2\, \pi_i(1-\pi_i)}{N_j}, \qquad \mathrm{Cov}(a_{ij}, a_{lj}) = -\frac{\bar{\nu}^2\, \pi_i \pi_l}{N_j}, \qquad i \neq l. \tag{10}$$

The covariance is *negative* for every pair of destinations sharing a source column, and this follows directly from the fact that a fixed number of histories are being split, exhaustively and mutually exclusively, among the possible destinations, so that more of N_j going to i automatically means

less going to l. Dividing the covariance by the geometric mean of the two variances, the common factor $\bar{\nu}^2/N_j$ cancels exactly, leaving a correlation that depends only on the destination probabilities themselves, not on N_j, which is a Serpent run-time setting, not a physical property of the fission process:

$$\rho_{\text{pred}}(i,l) = \frac{\text{Cov}(a_{ij}, a_{lj})}{\sqrt{\text{Var}(a_{ij})\,\text{Var}(a_{lj})}} = -\sqrt{\frac{\pi_i}{1-\pi_i} \cdot \frac{\pi_l}{1-\pi_l}}. \tag{11}$$

Equation (11) is written in terms of the destination probabilities π_i, which are not directly tallied; what Serpent reports is the mean coefficient $a_i = \bar{\nu}\pi_i$ (the expected number of secondaries reaching i per source neutron, i.e., $\pi_i = -\kappa a_i$ with $\kappa = -1/\bar{\nu}$). Substituting $\pi_i = -\kappa a_i$ into Eq. (11) and simplifying gives the closed form used for method (ii),

$$\rho_{\text{pred}}(i,l) = \kappa \frac{\sqrt{a_i a_l}}{\sqrt{(1+\kappa a_i)(1+\kappa a_l)}}, \qquad \kappa = -\frac{1}{\bar{\nu}}, \tag{12}$$

which requires only $\bar{\nu}$ (from `NUBAR` in `_res.m`) and the reference FM's own mean values – no additional MC replicas. (The multinomial variance and covariance results used in Eqs. (9-10) are standard; see, e.g., Ref. [12].)

Two simplifications in this derivation deserve a comment. The multiplicity $\bar{\nu}$ has been treated as a single constant, whereas ν depends on the incident neutron energy and each cell sees a slightly different spectrum. Each history has also been given the same weight, whereas Monte Carlo weights are not exactly unity.

Neither affects the amplitude of the perturbation. The matrix $\mathbf{D}$ of Eq. (6) is read from the statistical errors Serpent reports for each coefficient, and those already contain the local spectrum and the weights of the actual simulation. The multiplicity enters only through the conversion $\pi_i = a_i/\bar{\nu}$ needed to evaluate Eq. (12), because the common factor $\bar{\nu}^2/N_j$ cancels in Eq. (11). It is also worth noting that the multiplicity entering the tally is an expected value: Serpent increments the fission-matrix bin by the neutron weight times $\bar{\nu}(E_0)$, evaluated at the incident energy of the neutron inducing that fission, and the integer number of secondaries sent to the fission bank never enters the score. The form $X_s^{(i)} = \bar{\nu}\, I_s(i)$ therefore describes the estimator itself, and the variance of ν around its mean does not contribute to $\text{Var}(a_{ij})$.

Both simplifications enter the derivation at the same place. Neither changes the covariance of Eq. (10), which vanishes term by term as long as a history fissions once at most, whatever weight

and multiplicity it carries; both change only the variance that normalizes it. Writing c_i^2 for the relative variance of the per-event score $w\,\bar{\nu}(E)$, conditional on a fission in i, Eq. (11) becomes

$$\rho_{\mathrm{pred}}(i,l) = -\sqrt{\frac{\pi_i}{1-\pi_i+c_i^2}\cdot\frac{\pi_l}{1-\pi_l+c_l^2}}\,, \tag{13}$$

and the model used here is the case $c_i^2 = 0$. For the present calculations c^2 is small. The energy dependence of $\bar{\nu}$ contributes 1.5×10^{-4}, obtained from the eleven-group $\bar{\nu}$, fission cross section and flux of the reference calculation, and corresponding to a 1.24% RMS spread of $\bar{\nu}$ across the fission spectrum. The weights contribute a comparable amount, since they depart from unity only through the implicit treatment of (n,xn) reactions, which here accounts for 1.2×10^{-4} of the total source weight. Propagating $c^2 = 1.5\times10^{-4}$ through the full pipeline changes σ_{interp} by 0.004% on k_{eff} and by 0.002% on the fission-source RMS. Assigning conservatively the whole 1.24% spread as an error on the $\bar{\nu}$ of Eq. (12) changes them by 0.33% and 0.14%; for reference, the core-average $\bar{\nu}$ moves by 0.018% over the whole 300–1000 K database. The aim of Eq. (12) is to provide a justified definition of the correlation rather than an exact model of it, and these effects are well below the accuracy it demonstrates against the brute-force reconstruction in Sec. IV.B.

III.E.1. Uncorrelated treatment of Coefficients in Different Columns

The derivation above does not apply to coefficients that do *not* share a column, a_{ij} and $a_{kj'}$ with $j \neq j'$. It would be convenient, but not quite correct, to argue that N_j and $N_{j'}$ are simply independent: the total population $N = \sum_j N_j$ is fixed by the simulation's particle count per cycle, so source regions compete for a shared budget and $\mathrm{Cov}(N_j, N_{j'}) \neq 0$ in general. This coupling, however, need not propagate to the *normalized* coefficients themselves. By the law of total covariance,

$$\mathrm{Cov}(a_{ij}, a_{kj'}) = \underbrace{E\big[\,\mathrm{Cov}(a_{ij}, a_{kj'} \mid N_j, N_{j'})\big]}_{\text{(I)}} + \underbrace{\mathrm{Cov}\left(E[a_{ij} \mid N_j],\, E[a_{kj'} \mid N_{j'}]\right)}_{\text{(II)}}. \tag{14}$$

where term (I) vanishes because histories born in the disjoint columns j and j' are conditionally independent given $N_j, N_{j'}$ (again, no history-history interaction is assumed). Term (II) vanishes because $E[a_{ij} \mid N_j] = \pi_i$ is, by construction of the normalized estimator of Eq. (8), a constant that does not depend on the realized value of N_j, so its covariance with anything is zero. This

rules out one specific channel, competition for a shared per-cycle particle budget, as a source of cross-column correlation; it is not a claim that cross-column correlation is exactly zero for every other conceivable reason (e.g., spatially correlated local flux fluctuations, or the cycle-to-cycle propagation of the fission source itself [13]), and a bootstrap check on 150 independent replicas at 300 K does find a small residual: coefficient pairs sharing a column are negative 59.95% of the time (95% CI $[59.48\%, 60.30\%]$), against 51.13% $[50.96\%, 51.43\%]$ for pairs sharing a destination row and 51.20% $[51.11\%, 51.37\%]$ for pairs sharing neither — both of the latter statistically distinguishable from the 50% null, but by less than a tenth of the same-column effect. Method (ii) neglects this residual, as a simplifying assumption, since such residual is small relative to the effect being modeled. Its practical consequence is going to be illustrated in the result section, where method (ii) is compared against a genuine brute-force reconstruction that makes no such assumption.

III.E.2. Temperature Invariance of the Measured Correlation Matrix

Method (iii) assumes that the correlation matrix measured at 300 K also applies at the other database temperatures. The simplest test of this assumption is to measure $\boldsymbol{\rho}$ independently at the opposite end of the database range, 1000 K, and ask whether the two matrices differ by more than the statistical noise inherent in estimating either one.

Differences between correlation matrices are quantified here through the Frobenius norm, which treats an $N \times N$ matrix as a single vector of its entries and measures its Euclidean length,

$$\|\mathbf{M}\|_F = \sqrt{\sum_{a=1}^{N}\sum_{b=1}^{N} M_{ab}^2}\,. \tag{15}$$

The distance between two matrices is the norm of their difference; normalizing by the norm of the reference matrix gives the dimensionless *relative* Frobenius distance used throughout,

$$d_F(\mathbf{A},\mathbf{B}) = \frac{\|\mathbf{A}-\mathbf{B}\|_F}{\|\mathbf{B}\|_F}\,, \tag{16}$$

so that $d_F = 0$ for identical matrices and $d_F \sim 1$ for matrices differing by an amount comparable to their own magnitude.

Table I reports this distance between the 300 K and 1000 K matrices, alongside an internal noise floor for each temperature: the distance between two estimates of the *same* matrix, ob-

tained by splitting that temperature's own cycle data into two halves and measuring $\boldsymbol{\rho}$ from each. The noise floor is the distance one would obtain even if the true correlation matrix was exactly temperature-invariant, arising purely from the finite statistics of the estimate.

TABLE I
Simple Check of Temperature Invariance for the Measured Correlation Matrix

Comparison	Relative Frobenius distance
$\boldsymbol{\rho}$(300 K) vs. $\boldsymbol{\rho}$(1000 K)	0.643
$\boldsymbol{\rho}$(300 K), split-half internal noise floor	0.909
$\boldsymbol{\rho}$(1000 K), split-half internal noise floor	0.909

The 300 K vs. 1000 K distance is *smaller* than the noise floor of estimating $\boldsymbol{\rho}$ from half the data at a single, fixed temperature. By this measure we can state that the correlation matrix changes less between the two temperature extremes than a single-temperature estimate fluctuates under its own finite statistics, so reusing one matrix across the whole database is at least as reasonable as trusting the estimate at any one temperature on its own.

III.F. Definition of error metrics

Two plain metrics are used throughout, deliberately without a covariance-weighted quadratic form such as a Mahalanobis distance: they are less sensitive to the *spatial pattern* of a residual, but directly interpretable, which is what matters for our case-by-case comparison. For k_{eff},

$$\Delta k_{\text{eff}} = k_{\text{eff}}^{\text{interp}} - k_{\text{eff}}^{\text{ref}}, \qquad z = \frac{\Delta k_{\text{eff}}}{\sigma_{\text{tot}}}. \tag{17}$$

The fission source F is normalized to unit sum over the 127 cells giving $\tilde{F}_i$. Its local relative deviation in each zone is defined as

$$\delta\tilde{F}_i = \frac{\tilde{F}_i^{\,\text{interp}} - \tilde{F}_i^{\,\text{ref}}}{\tilde{F}_i^{\,\text{ref}}} \times 100\%, \tag{18}$$

and the overall discrepancy between the interpolated and reference source distributions is summarized by the root-mean-square (RMS) of the zone-wise relative deviations,

$$\text{RMS} = \sqrt{\left\langle \delta\tilde{F}_i^{\,2} \right\rangle_i}. \tag{19}$$

The RMS therefore provides a single scalar measure of the average relative error over the entire fission source distribution, without accounting for the spatial arrangement of the residuals.

Every uncertainty band reported in Sec. IV is $\pm 2\sigma_{\text{tot}}$, matching the $|z| < 2$ threshold used as the "consistent with noise" criterion, so that falling inside or outside a plotted band coincides with passing or failing that test. σ_{database} in Eq. (5) uses the analytical model (ii) throughout Sec. IV, computationally validated next.

IV. RESULTS

The pipeline is applied to twelve validation cases, each featured by a different in-core temperature field imposed on the HPMR model, reported in Figure 3 and Figure 4. They are chosen to span a wide range of spatial sharpness, since it is precisely under sharp gradients that the local-temperature interpolation assumption – and hence the CTCR – is most severely tested. The set comprises two spatially sharp "Dirac-like" perturbations, in which a single central assembly is offset in temperature from an otherwise uniform core; a six-assembly and two ring-shaped hot-spot patterns; a smooth linear gradient across the core; and six power-shaped profiles of varying peak sharpness and radial offset. Each case is evaluated in both the *raw* (interpolation only) and *corrected* (interpolation plus CTCR) variants against a direct Serpent reference calculation of the same temperature field.

IV.A. Reference Case Uncertainty

Figure 5 shows the reference case's uncertainty σ_{ref} (see Eq. 5). Averaged over the twelve cases, it is 9.7 pcm on k_{eff} and 0.309% (RMS) on the fission source, and is essentially uniform across cases; the one mild outlier is the power shape 300–1000 K case (12.5 pcm, 0.42%), whose broader temperature profile produces a less peaked fission-source distribution and hence a statistically noisier detector tally. This is only half of the noise budget of Eq. (5): on its own it says nothing about the uncertainty introduced by the interpolation, quantified next.

IV.B. Validating the Database-Noise Models Against Brute Force

Figure 6 compares the three inexpensive approaches: uncorrelated (*i*), analytical (*ii*), and single-seed measured (*iii*), against the brute-force reconstruction (*iv*), for the corrected variant of

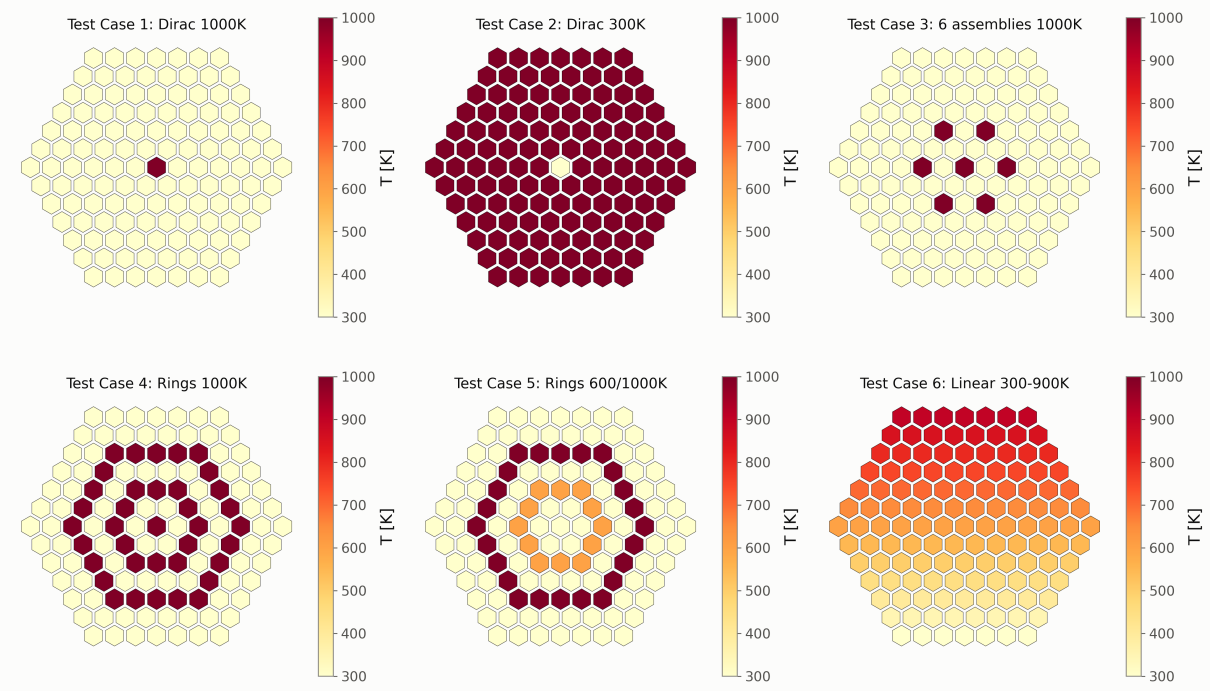

Fig. 3. Reference-cases temperature distribution (part 1).

all twelve cases, on both observables. Agreement is summarized by the mean absolute deviation from brute force across the twelve cases. The ordering of the interpolated uncertainties σ_{interp} predicted by the three models, excluding the brute-force results, which are used here as reference, is consistent with expectations in almost all the test cases. The uncorrelated model systematically yields the largest σ_{interp}, and therefore the widest confidence intervals, the only exception being k_{eff} in Test Case 4; the analytical covariance model gives the smallest values, while the covariance matrix obtained by considering every cycle of the MC simulation, lies in between. This single exception can be ascribed to statistical noise: the database is resampled randomly from a Gaussian distribution centered on the nominal values, so fluctuations of this magnitude are not unexpected.

F k_{eff}, the analytical model is the most accurate of the three, reproducing the brute-force uncertainty to within 4.6%, ahead of the single-seed computed model (8.7%) and well ahead of the uncorrelated model (18.0%). It can be noticed that the intra-column correlations are negative, so neglecting them (method i) inflates the propagated k_{eff} variance, while capturing them in closed form (method ii) does not require the additional simulation that measuring them (method iii) does. On the fission-source RMS the ranking changes: the single-seed (3.3%) and the analytic

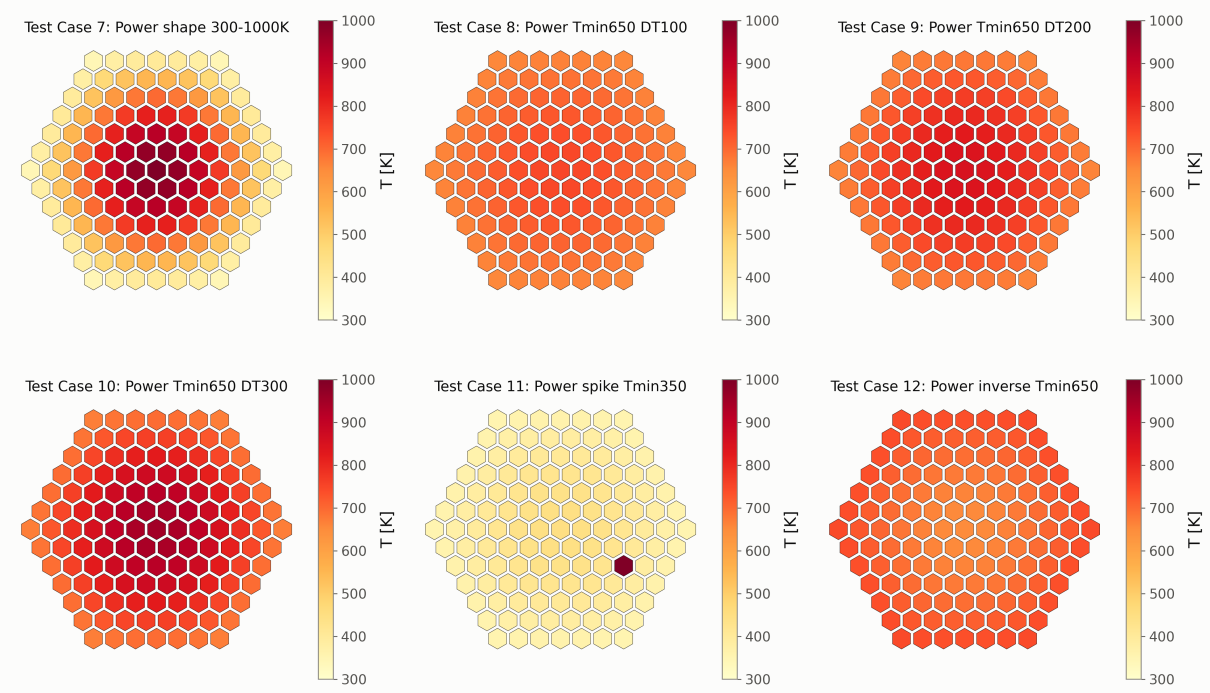

Fig. 4. Reference-cases temperature distribution (part 2).

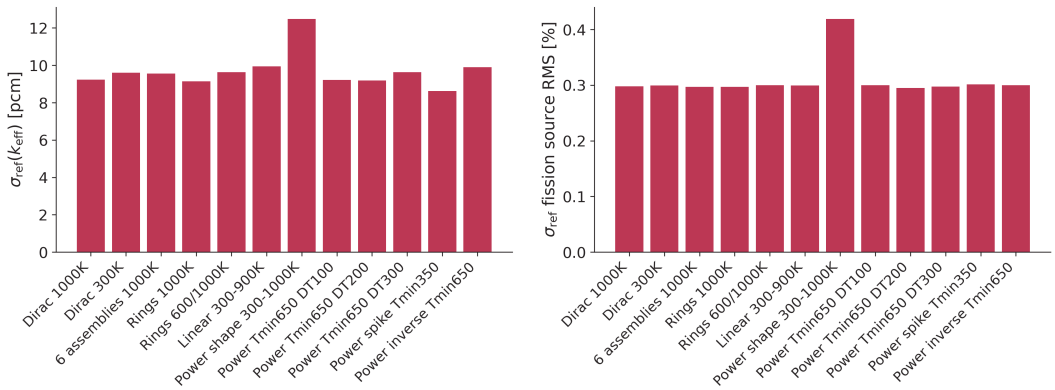

Fig. 5. Reference-case Monte Carlo uncertainty σ_{ref}: k_{eff} (left) and fission-source RMS (right), for all twelve cases.

(3.6%) models track brute force more closely than the uncorrelated one (7.3%). This is consistent with the cross-column residual left unmodeled by method (ii): the fission source is a less global quantity, sensitive to correlations between coefficients in different columns in a way the eigenvalue is not, so the very terms the analytical model omits matter more here.

In the following sections, the analytical model and the brute-force method are adopted from

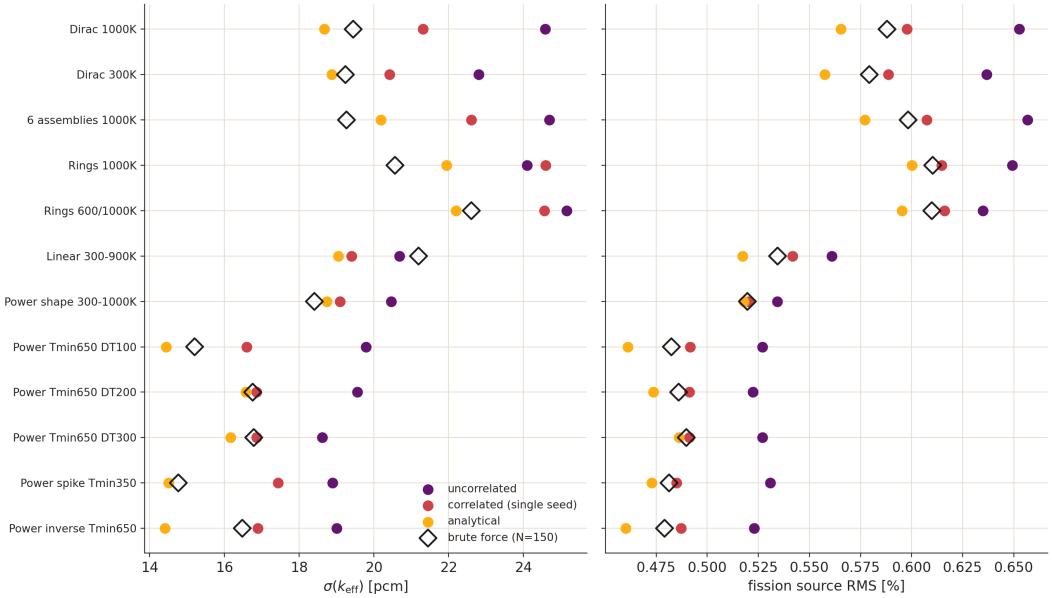

Fig. 6. Database-side uncertainty predicted by the different σ_{interp} quantification methods against the brute-force reference (iv, $N = 150$): k_{eff} (left) and fission-source RMS (right), corrected variant, all twelve cases.

among the four uncertainty propagation methods considered. The selection of the analytical model is not driven solely by the accuracy of the results with respect to the uncertainty obtained with the brute-force method, since all four methods provide very similar results, but also by its simplicity of application. Moreover, the analytical model allows for a conservative approach: as shown in Figure 6, it provides the lowest uncertainty values. When combined with the uncertainty of the reference results, this leads to the smallest total uncertainty budget and, consequently, to the narrowest acceptance window for the results.

IV.B.1. Stability of the Analytical Model

It is reasonable to object that the analytical model is not truly independent of Monte Carlo noise: the closed form of Eq. (12) is evaluated on the mean coefficients a_i of a single database, which are themselves affected by statistical uncertainty. If a different random seed was used, the predicted matrix $\boldsymbol{\rho}_{\text{pred}}$ would differ slightly, and one may ask whether that sensitivity matters.

To test this, $\boldsymbol{\rho}_{\text{pred}}$ at 300 K was rebuilt from each of the 150 independent seeds of the brute-force set and compared, through the relative Frobenius distance of Eq. (16), against the matrix built from the production seed (Table II).

TABLE II
Sensitivity of $\boldsymbol{\rho}_{\text{pred}}$ to the Database Seed (300 K)

Seed-to-seed spread	Relative Frobenius distance
Mean	0.0013
[16, 84] percentile	[0.0013, 0.0013]
Maximum	0.0014

The Monte Carlo noise in the mean FM coefficients is therefore not a meaningful source of uncertainty for the analytical model, and $\boldsymbol{\rho}_{\text{pred}}$ can be treated as effectively deterministic.

IV.C. Uncorrected and Corrected Results Comparison

Figure 7 compares raw and corrected interpolation against the direct reference, before any uncertainty is folded in, and Table III reports the underlying per-case numbers. Averaged over the twelve cases, the CTCR reduces the mean absolute k_{eff} discrepancy from 40.8 to 23.7 pcm (42%) and the mean fission-source RMS discrepancy from 1.27% to 0.94% (26%). The benefit is markedly uneven. It is largest for the sharpest, most localized profiles — Rings 1000 K improves from 57.4 to 1.2 pcm, and 6 assemblies 1000 K from 53.9 to 24.0 pcm — consistent with the CTCR having been calibrated against a sharp central perturbation. For smoother, less localized profiles it is negligible or mildly unfavorable, see e.g. the Power Tmin650 DT200 case worsening from −41.5 to −52.8 pcm. This geometric dependence is expected by design of the correction: as it is calibrated on a Dirac-like perturbation, it tends to be more effective when the temperature field realisation is closer to the calibration profile.

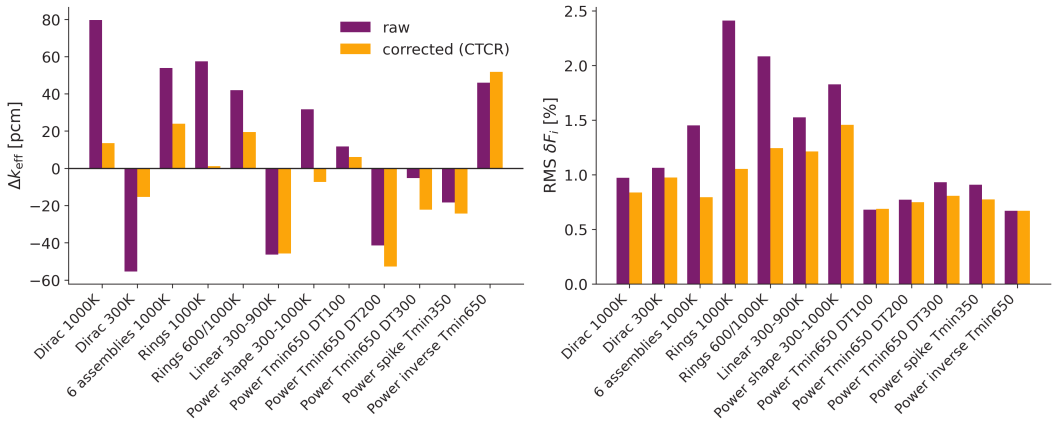

Fig. 7. k_{eff} discrepancy (left) and fission-source RMS discrepancy (right), raw vs. CTCR-corrected interpolation, all twelve cases, with no uncertainty band.

TABLE III
Point-Estimate Discrepancy Against the Direct Reference, All Twelve Cases

	Δk_{eff} [pcm]		Fission-source RMS [%]	
Case	raw	corrected	raw	corrected
Dirac 1000 K	79.7	13.5	0.972	0.838
Dirac 300 K	−55.4	−15.4	1.064	0.975
6 assemblies 1000 K	53.9	24.0	1.451	0.794
Rings 1000 K	57.4	1.2	2.411	1.052
Rings 600/1000 K	42.0	19.5	2.084	1.244
Linear 300–900 K	−46.3	−45.8	1.525	1.212
Power shape 300–1000 K	31.7	−7.4	1.828	1.456
Power Tmin650 DT100	11.7	6.0	0.680	0.688
Power Tmin650 DT200	−41.5	−52.8	0.772	0.748
Power Tmin650 DT300	−5.2	−22.2	0.932	0.806
Power spike Tmin350	−18.4	−24.3	0.909	0.775
Power inverse Tmin650	46.0	51.8	0.671	0.670
Mean $\lvert\cdot\rvert$	40.8	23.7	1.275	0.938

IV.D. Reuslts with the Combined Uncertainty

This section proposes the same results of the previous one but with a deeper look the the uncertainty budget and thus the results acceptability.

Combining σ_{ref} with σ_{interp} from the analytical model in quadrature gives the total uncertainty σ_{tot} of Eq. (5). Under this combined budget, 58% of the corrected cases fall within $|z| < 2$ (median $|z| = 1.30$), against 42% of the raw cases (median $|z| = 2.18$): a consistent and, for the corrected variant, now a majority of cases. σ_{interp} is 14–22 pcm against 9–12 pcm for σ_{ref} in every case, so the database term still dominates the budget and both variants are judged against a relatively wide band.

Rather than collapsing this budget to a single $\pm 2\sigma_{\text{tot}}$ number per case, Figure 8 shows the full picture for k_{eff}: every one of the $N = 500$ analytical-model realizations (gray) and, since 150 genuinely independent brute-force replicas of the whole database already exist, every one of those true replicas too (blue open circles), alongside the reference case's own $\pm 2\sigma_{\text{ref}}$ (pink band) and the raw/corrected point estimates (black/red).

For the fission source (Fig. 10) the same full-distribution logic is applied but this time, only the brute force method cell-wise result is shown, because the $N = 500$ synthetic analytical realizations, of every F_i, add visual clutter without adding information beyond what is already displayed in 6, so only the $N = 150$ independent brute-force replicas are shown, split by flavor: raw

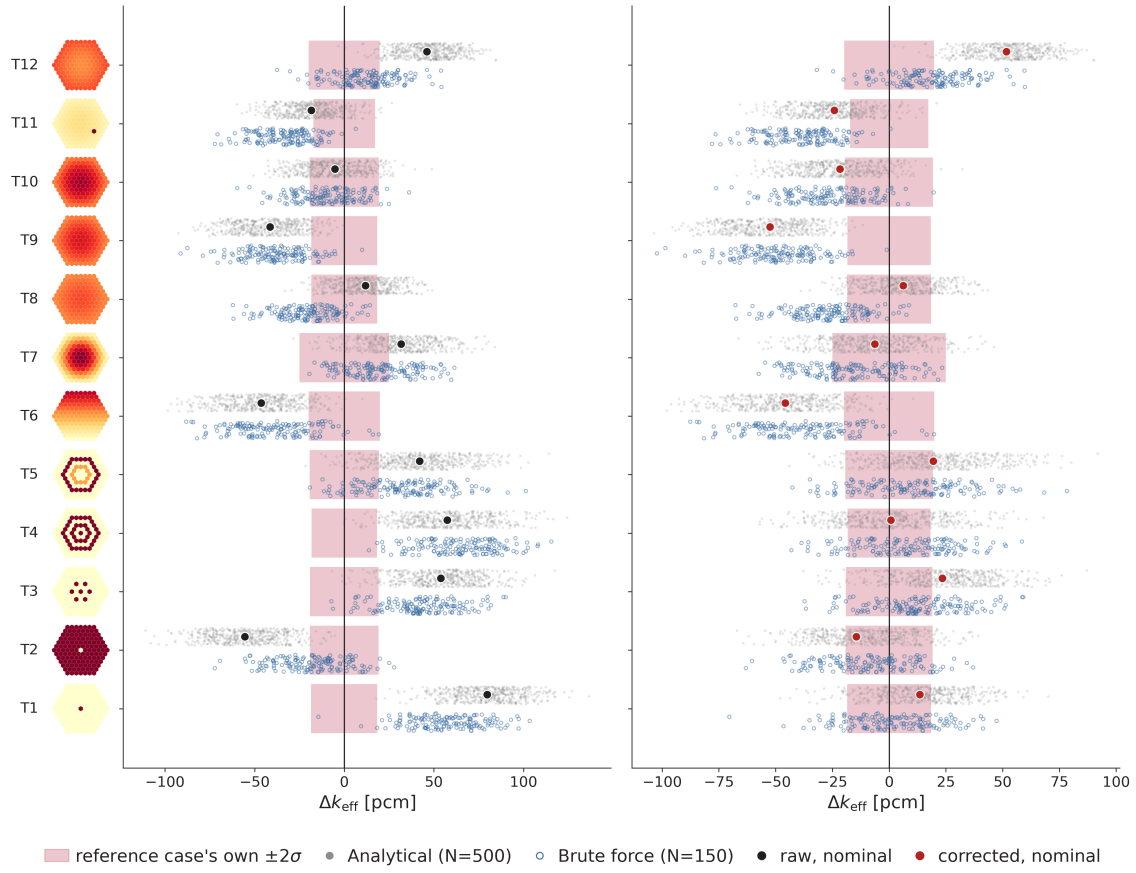

Fig. 8. k_{eff} discrepancy, raw (left) and corrected (right): every analytical-model realization (gray, $N = 500$) and every true independent brute-force replica (blue open circles, $N = 150$), with the reference case's own $\pm 2\sigma_{\text{ref}}$ (pink band) and the point estimate (filled circle), all twelve cases.

in gray, corrected in shades of red. A corrected point is recolored gold if and only if it falls inside the σ_{ref} band *and* the matching raw point, of the same zone, was outside it. Every other corrected point — already consistent with noise in raw, or still not explained by noise even after correction — keeps the base red. This isolates the population the correction is meant to act on, rather than diluting the plot with every in-band point regardless of whether raw needed fixing there at all. Figure 9 reports the core assemblies numbering to better navigate the results in Fig. 10 where only the first six test cases are displayed for sake of shortness, but still well displays both cases, where the CTCR has an higher impact on local quantity F_i, and where temperature differences are less sharp and so the CRCT less needed. The remaining six cases are reported in 11).

This systematic offset of the whole replica distribution away from zero is itself informative: it

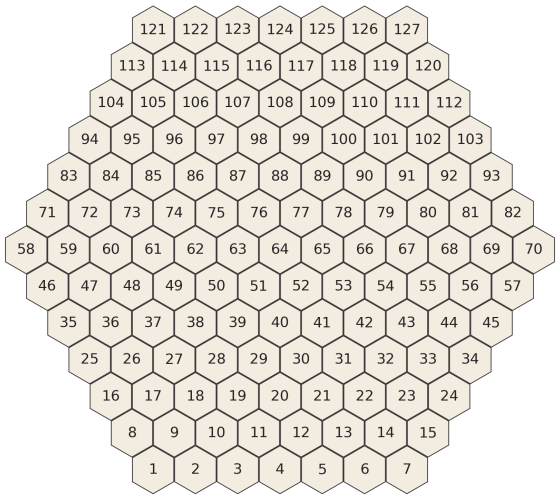

Fig. 9. Hexagon-index map (1–127) referenced by the zone-index axis of Fig. 10 and Fig. 11; hexagon 64 is the central assembly C of the Dirac calibration cases.

points to a mismatch between the database and some of the test cases that no amount of resampling the database can correct, because it is not database noise at all. The database is tallied with the radial reflector at the same uniform temperature as the rest of the core at each of its eight reference points, whereas several test cases instead fix the reflector at a constant 300 K regardless of the in-core field being tested. Where the in-core temperature departs from 300 K, this creates a sharp, systematic core-to-reflector gradient that is simply absent from the database's own construction, and every replica of that database — brute-force or synthetic — inherits the same mismatch, hence the same offset. This is a bias in the comparison's reference frame, not a statistical fluctuation.

Where the corrected replica cloud is centered close to zero — Dirac 300 K, Linear 300–900 K, and most of the smoother power-shaped profiles — the residual deviation from the reference is statistically indistinguishable from Monte Carlo noise (and the reflector-mismatch bias just described, where present, is small relative to that noise), and no further correction is warranted. Where it is not — the localized spikes in Dirac 1000 K, 6-assemblies 1000 K, and the Rings cases — the deviation exceeds what either noise source can account for, and it therefore indicates an uncorrected bias rather than statistical scatter. Because the database-noise model has itself been validated against brute force, it is not the plausible origin of these residuals; the most likely cause is the interpolation scheme's finite temperature grid, whose eight-point spacing cannot fully resolve the sharpest gradients. Figure 9 maps each zone index in Fig. 10 and back to its physical hexagon.

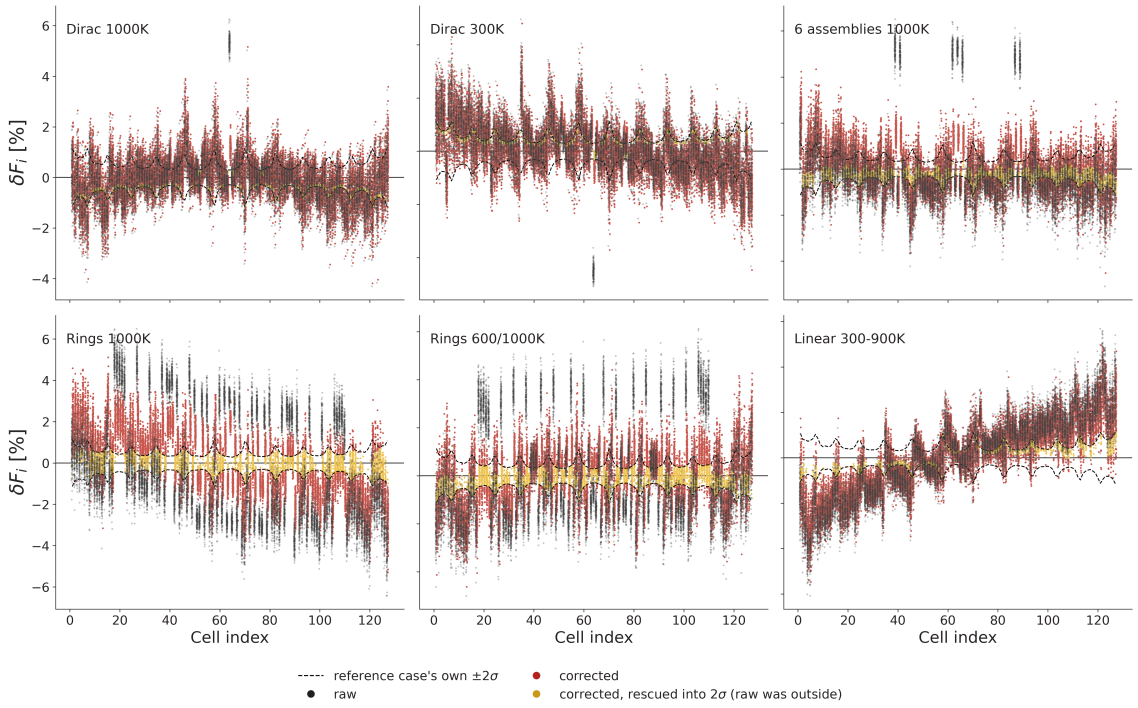

Fig. 10. δF_i of the raw (red dots) and corrected (black dots) cases. The softer colored dots represent the dispersion of δF_i using the brute-force method only. Red points are turned gold only when are inside the $\pm 2\sigma_{\text{ref}}$ band and any of the black dots of the same region are.

A global summary metric can be misleading about how the correction is actually performing locally. It has been already reported the CTCR's benefit on the two aggregated figures of merit, k_{eff} and the fission-source RMS: both improve, but a global metric averages over all 127 zones, and so cannot distinguish a case that improves a little everywhere from one that improves dramatically in exactly the zone that matters and not at all elsewhere. Table IV makes the zone-by-zone picture concrete on a per-case basis: for each of the twelve cases, the number of zones (out of 127) whose fission-source point estimate is statistically consistent with noise ($|\delta F_i| < 2\sigma_{\text{tot}}$), raw versus corrected, under each of the four σ_{inerp} models.

The pattern is consistent with the point-estimate result of Table III, but is sharper here: for the Rings cases the corrected count roughly doubles (e.g., Rings 1000 K rises from 36–41 to 77–88 zones out of 127, consistently across all four models), while for the smoothest profiles the shift is small or even mildly unfavorable (Dirac 300 K, Power Tmin650 DT100/DT200). The Appendix collects the remaining six cases; among them Power shape 300–1000 K shows one of the largest improvements of the whole set, its corrected count rising from 59 to 76 zones (Fig. 11). It

TABLE IV
Number of Zones (out of 127) Statistically Consistent with Noise ($|\delta F_i| < 2\sigma_{\text{tot}}$), Raw (r) vs. Corrected (c), by Case and σ_{database} Model

	(i) Uncorr.		(ii) Analyt.		(iii) Meas.		(iv) Brute force	
Case	r	c	r	c	r	c	r	c
Dirac 1000 K	116	118	110	109	112	112	113	115
Dirac 300 K	110	106	103	100	105	100	107	105
6 assemblies 1000 K	103	114	102	103	103	106	103	112
Rings 1000 K	41	84	36	77	37	79	40	88
Rings 600/1000 K	55	87	51	85	55	85	55	88
Linear 300–900 K	66	90	62	82	65	88	66	89
Power shape 300–1000 K	61	77	59	76	60	76	60	76
Power Tmin650 DT100	118	116	115	114	116	114	117	115
Power Tmin650 DT200	117	119	113	113	114	114	115	115
Power Tmin650 DT300	97	104	96	101	93	102	97	103
Power spike Tmin350	109	109	101	106	103	107	107	107
Power inverse Tmin650	120	119	114	115	117	116	117	116

is important to emphasize the accurate reproduction of local quantities, since in many transient analyses the quantity of interest is not a core-averaged parameter but rather the most critical point in the system, such as a localized hot spot. In these situations, the reliability of the interpolated model is determined by its ability to accurately predict the behavior at that specific location, rather than by the overall agreement in terms of global quantities such as k_{eff} or the RMS error. Indeed, it is the accuracy at the most critical zone—where an error would have the greatest impact—that ultimately determines whether the model can be trusted.

V. CONCLUSIONS

The interpolated and CTCR-corrected fission matrix inherits statistical noise from the database it is built from, but — unlike a direct MC calculation — its resulting fundamental eigenvalue and eigenfunction has no uncertainty of its own to read off from any single simulation output; it has to be constructed by resampling the database itself. Of the three cheap ways of doing this, a zero-replica analytical model, derived from the branching structure of the fission process under the explicit simplifying assumption that only same-column coefficients are correlated, reproduces a $N = 150$ brute-force reconstruction's k_{eff} uncertainty at least as well as an approach that requires measuring a full correlation matrix, and both provides markedly on assuming no correlation at all; on the fission-source RMS specifically, the two correlated approaches perform comparably to each other and somewhat better than the analytical model, though all three remain within the same order of magnitude of brute force. Folding either uncertainty source into a single budget,

the CTCR's point-estimate benefit persists once both noise sources are accounted for, and most of the remaining case-by-case discrepancies are statistically consistent with this combined floor. Where they are not, the residual is plausibly a limitation of the interpolation scheme's finite temperature grid rather than of the fission-matrix noise model used to characterize it — a distinction this budget makes possible, and a natural target for future refinement independent of any further statistical work on the covariance itself. Overall the average error on the k_{eff} and on the RMS of the fission source, considering all the test cases of the raw cases are, respectively, 40.8 pcm and 1.27%. After the CTCR is applied, the error of the two parameters reduces to 23.7 pcm and 0.94%.

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APPENDIX: FISSION-SOURCE DEVIATION, TEST CASES 7–12

Fig. 10 in the main text shows the six test cases containing the most localized temperature perturbations, the regime for which the CTCR is calibrated. Fig. 11 gives the same plot, in the same style, for the remaining six cases, which form the power-shaped family.

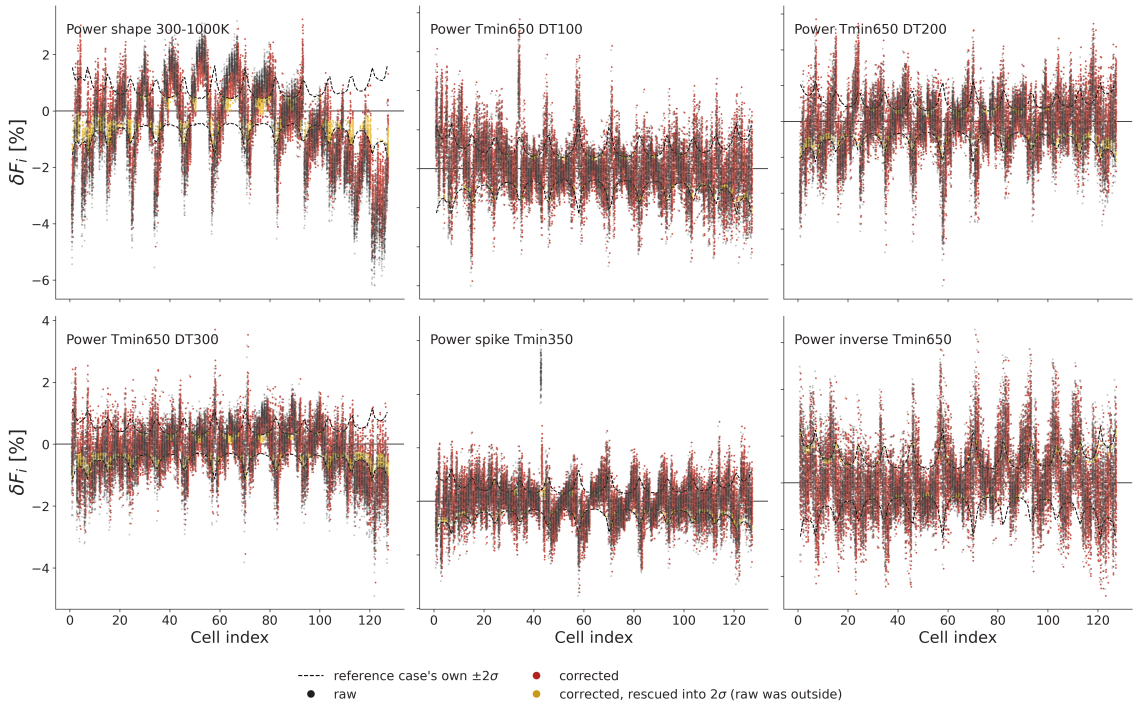

Fig. 11. As Fig. 10, for test cases 7–12.

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